

● HARD

● ALGEBRA

Systems of two linear equations in two variables

03

10 questions · ~15 min

Q1

ALGEBRA

During a month, Morgan ran r miles at 5 miles per hour and biked b miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

- A. 80
- B. 100
- C. 120
- D. 160

Q2

GRID-IN ALGEBRA

$$5y = 10x + 11$$

$$-5y = 5x - 21$$

The solution to the given system of equations is x, y . What is the value of $30x$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$2x + 3y = 7$$

$$10x + 15y = 35$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

A. $\frac{r}{5} + 7, -\frac{r}{5} + 35$

B. $-\frac{3r}{2} + \frac{7}{2}, r$

C. $r, \frac{2r}{3} + \frac{7}{3}$

D. $r, -\frac{3r}{2} + \frac{7}{2}$

A sample of a certain alloy has a total mass of 50.0 grams and is 50.0% silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. What was the mass, in grams, of the silicon in the second piece?

- A. 9.0
- B. 16.0
- C. 20.0
- D. 30.0

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

A. $r, -\frac{6r}{7} + \frac{5}{7}$

B. $r, \frac{7r}{6} + \frac{5}{6}$

C. $\frac{r}{4} + 5, -\frac{r}{4} + 20$

D. $-\frac{6r}{7} + \frac{5}{7}, r$

Q7

GRID-IN ALGEBRA

A piece of wire with a length of 32 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 4 more than 3 times the value of y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN ALGEBRA

$$24x + y = 48$$

$$6x + y = 72$$

The solution to the given system of equations is (x, y) . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ALGEBRA

$$48x - 64y = 48y + 24$$

$$ry = \frac{1}{8} - 12x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN ALGEBRA

$$-x + y = -3.5$$

$$x + 3y = 9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.